

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report:
Locating Log Files

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Objectives

- Get familiar with the structure and content of Linux log files.
- Locate log files in an unfamiliar Linux system.
- Monitor log files in real time using tail and journalctl.

Part 1 – Log File Overview

Step 1a – Apache Web Server Log Entry Analysis

The following single log entry was examined:

```
[Wed Mar 22 11:23:12.207022 2017] [core:error] [pid 3548:tid 4682351596]  
[client 209.165.200.230] File does not exist:  
/var/www/apache/htdocs/favicon.ico
```

Field	Value	Meaning
Timestamp	Wed Mar 22 11:23:12.207022 2017	Date and time the event occurred
Type	core:error	An error-level event from the core Apache module
PID / TID	3548 / 4682351596	Process ID and Thread ID handling the request
Client IP	209.165.200.230	IP address of the requesting client
Description	File does not exist: /var/www/apache/htdocs/favicon.ico	The browser requested favicon.ico but the file was not found on the server

Answer – What happened?

On Wednesday, March 22, 2017 at 11:23:12, the Apache web server received a request from client 209.165.200.230 for the file /var/www/apache/htdocs/favicon.ico. Apache could not find the file on disk and logged a core:error event. This is a common, non-critical error that occurs when a browser automatically requests a site's favicon and the file has not been placed in the web root directory.

Step 1b – cat /var/log/logstash-tutorial.log

```
[analyst@secOps ~]$ cat /var/log/logstash-tutorial.log
```

```
81.220.24.207 - - [04/Jan/2015:05:24:57 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/blog/geekery/ssl-latency.html" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.73.11" (KHTML, like Gecko) Version/7.0.1 Safari/537.73.11"
81.220.24.207 - - [04/Jan/2015:05:24:57 +0000] "GET /images/web/2009/banner.png HTTP/1.1" 200 52315 "http://www.semicomplete.com/blog/geekery/ssl-latency.html" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.73.11" (KHTML, like Gecko) Version/7.0.1 Safari/537.73.11"
81.220.24.207 - - [04/Jan/2015:05:24:58 +0000] "GET /favicon.ico HTTP/1.1" 200 3638 "http://www.semicomplete.com/blog/geekery/ssl-latency.html" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.73.11" (KHTML, like Gecko) Version/7.0.1 Safari/537.73.11"
66.249.73.135 - - [04/Jan/2015:05:25:05 +0000] "GET /blog/geekery/vmware-cpu-performance.html HTTP/1.1" 200 12908 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like Mac OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0 Mobile/10A5376e Safari/8536.25 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)"
46.105.14.53 - - [04/Jan/2015:05:26:17 +0000] "GET /blog/tags/puppet?flav=rss20 HTTP/1.1" 200 14872 "-" "UniversalFeedParser/4.2-pre-314-svn +http://feedparser.org/"
218.30.103.62 - - [04/Jan/2015:05:27:05 +0000] "GET /robots.txt HTTP/1.1" 200 - "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:27:10 +0000] "GET /robots.txt HTTP/1.1" 200 - "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:27:15 +0000] "GET /projects/fex/ HTTP/1.1" 200 14352 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
74.125.40.20 - - [04/Jan/2015:05:27:22 +0000] "GET /?flav=rss20 HTTP/1.1" 200 29941 "-" "FeedBurner/1.0 (http://www.FeedBurner.com)"
71.212.224.97 - - [04/Jan/2015:05:27:34 +0000] "GET /projects/xdotool/ HTTP/1.1" 200 12292 "http://suckless.org/rocks" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:34 +0000] "GET /reset.css HTTP/1.1" 200 1015 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:35 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:35 +0000] "GET /images/jordan-80.png HTTP/1.1" 200 6146 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
71.212.224.97 - - [04/Jan/2015:05:27:35 +0000] "GET /images/web/2009/banner.png HTTP/1.1" 200 52315 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_2) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
218.30.103.62 - - [04/Jan/2015:05:27:36 +0000] "GET /projects/xdotool/xdotool.xhtml HTTP/1.1" 304 - "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
108.174.55.234 - - [04/Jan/2015:05:27:45 +0000] "GET /?flav=rss20 HTTP/1.1" 200 29941 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:27:57 +0000] "GET /blog/geekery/c-vs-python-bdb.html HTTP/1.1" 200 11388 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
121.107.188.202 - - [04/Jan/2015:05:27:57 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-dashboards3.png HTTP/1.1" 200 171717 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/32.0.1700.107 Safari/537.36"
218.30.103.62 - - [04/Jan/2015:05:28:21 +0000] "GET /blog/productivity/better-zsh-xterm-title-fix.html HTTP/1.1" 200 10185 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:28:43 +0000] "GET /blog/geekery/xvfb-firefox.html HTTP/1.1" 200 10975 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:29:06 +0000] "GET /blog/geekery/puppet-facts-into-mcollective.html HTTP/1.1" 200 9872 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
198.46.149.143 - - [04/Jan/2015:05:29:13 +0000] "GET /blog/geekery/disabling-battery-in-ubuntu-vms.html?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+semicomplete%2Fmain+%28semicomplete.com%29+Jordan+Sissel%29 HTTP/1.1" 200 9316 "-" "Tiny Tiny RSS/1.11 (http://tt-rss.org/)"
198.46.149.143 - - [04/Jan/2015:05:29:13 +0000] "GET /blog/geekery/solving-good-or-bad-problems.html?utm_source=feedburner&utm_medium=feed&utm_campaign=Feed%3A+semicomplete%2Fmain+%28semicomplete.com%29+Jordan+Sissel%29 HTTP/1.1" 200 10756 "-" "Tiny Tiny RSS/1.11 (http://tt-rss.org/)"
218.30.103.62 - - [04/Jan/2015:05:29:26 +0000] "GET /blog/geekery/jquery-interface-puffer.html%20target= HTTP/1.1" 200 202 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
218.30.103.62 - - [04/Jan/2015:05:29:48 +0000] "GET /blog/geekery/ec2-reserved-vs-ondemand.html HTTP/1.1" 200 11834 "-" "Sogou web spider/4.0(+http://www.sogou.com/docs/help/webmasters.htm#07)"
66.249.73.135 - - [04/Jan/2015:05:30:06 +0000] "GET /blog/web/firefox-scrolling-fix.html HTTP/1.1" 200 8950 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 like Mac OS X) AppleWebKit/536.26 (KHTML, like Gecko) Version/6.0 Mobile/10A5376e Safari/8536.25 (compatible; Googlebot/2.1; +http://www.google.com/bot.html)"
86.1.76.62 - - [04/Jan/2015:05:30:37 +0000] "GET /projects/xdotool/ HTTP/1.1" 200 12292 "http://www.haskell.org/haskellwiki/Xmonad/Frequently_asked_questions" "Mozilla/5.0 (X11; Linux x86_64; rv:24.0) Gecko/20140205 Firefox/24.0 Iceweasel/24.3.0"
86.1.76.62 - - [04/Jan/2015:05:30:37 +0000] "GET /reset.css HTTP/1.1" 200 1015 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (X11; Linux x86_64; rv:24.0) Gecko/20140205 Firefox/24.0 Iceweasel/24.3.0"
86.1.76.62 - - [04/Jan/2015:05:30:37 +0000] "GET /style2.css HTTP/1.1" 200 4877 "http://www.semicomplete.com/projects/xdotool/" "Mozilla/5.0 (X11; Linux x86_64; rv:24.0) Gecko/20140205 Firefox/24.0 Iceweasel/24.3.0"
[analyst@secOps ~]$
```

Answer – Is this still a web transaction? Why is the format different?

Yes, the entries in logstash-tutorial.log still represent web transactions — each line records an HTTP request made by a client, including the client IP address, timestamp, HTTP method, requested resource, response code, bytes transferred, referrer URL, and user-agent string.

The format appears different from the Apache error-log entry in item (a) because this file uses the Combined Log Format (a common access-log format), whereas the entry in (a) is from Apache's error log, which uses a distinct format to capture error-specific fields such as severity level, process/thread IDs, and a human-readable description. Access logs and error logs serve different purposes and therefore use different formats.

Step 2a – sudo more /var/log/messages

```
[analyst@secOps ~]$ sudo more /var/log/messages
[analyst@secOps ~]$ sudo more /var/log/messages
[sudo] password for analyst:
Mar 20 08:34:38 secOps kernel: [6.149910] random: crng init done
Mar 20 08:34:40 secOps kernel: [8.280667] floppy0: no floppy controllers found
Mar 20 08:34:40 secOps kernel: [8.280724] work still pending
Mar 20 08:35:16 secOps kernel: [ 44.414695] hrtimer: interrupt took 5346452 ns
Mar 20 14:28:29 secOps kernel: [21239.566409] pcnet32 0000:00:03.0 enp0s3: link down
Mar 20 14:28:33 secOps kernel: [21243.404646] pcnet32 0000:00:03.0 enp0s3: link up, 100Mbps,
full-duplex
Mar 20 14:28:35 secOps kernel: [21245.536961] pcnet32 0000:00:03.0 enp0s3: link down
Mar 20 14:28:43 secOps kernel: [21253.427459] pcnet32 0000:00:03.0 enp0s3: link up, 100Mbps,
full-duplex
Mar 20 14:28:53 secOps kernel: [21263.449480] pcnet32 0000:00:03.0 enp0s3: link down
Mar 20 14:28:57 secOps kernel: [21267.500152] pcnet32 0000:00:03.0 enp0s3: link up, 100Mbps,
full-duplex
Mar 20 14:29:01 secOps kernel: [21271.551499] pcnet32 0000:00:03.0 enp0s3: link down
Mar 20 14:29:05 secOps kernel: [21275.389707] pcnet32 0000:00:03.0 enp0s3: link up, 100Mbps,
full-duplex
Mar 22 06:01:40 secOps kernel: [0.000000] Linux version 4.8.12-2-ARCH (builduser@andyrtr)

(gcc version 6.2.1 20160830 (GCC) ) #1 SMP PREEMPT Fri Dec 2 20:41:47 CET 2016
Mar 22 06:01:40 secOps kernel: [0.000000] x86/fpu: Supporting XSAVE feature 0x001: 'x87
floating point registers'
Mar 22 06:01:40 secOps kernel: [0.000000] x86/fpu: Supporting XSAVE feature 0x002: 'SSE
registers'
Mar 22 06:01:40 secOps kernel: [0.000000] x86/fpu: Supporting XSAVE feature 0x004: 'AVX
registers'
Mar 22 06:01:40 secOps kernel: [0.000000] x86/fpu: xstate_offset[2]: 576, xstate_sizes[2]:
256
Mar 22 06:01:40 secOps kernel: [0.000000] x86/fpu: Enabled xstate features 0x7, context size
is 832 bytes, using 'standard' format.
Mar 22 06:01:40 secOps kernel: [0.000000] x86/fpu: Using 'eager' FPU context switches.
<some output omitted>
```

Answer – Evidence of slow network around 4:20 am on May 19?

Based on the log entries provided in the lab, there is no direct evidence of a slow network event at 4:20 am on May 19. The sample output shown covers dates in March, not May 19.

However, the entries from Mar 20 14:28:29 – 14:29:05 demonstrate a pattern of repeated link-down / link-up cycling on interface enp0s3 (the network adapter), which would cause intermittent connectivity and appear as slowness to users. If similar link-flapping entries existed around 4:20 am on May 19, those would constitute evidence of the reported network issue. To investigate, one would search the log for entries timestamped 'May 19' between 04:15 and 04:25 containing 'link down' or 'link up'.

Part 2 – Locating Log Files in Unknown Systems

Step a – man nginx

```
[analyst@secOps ~]$ man nginx
```

```
NGINX(8)                BSD System Manager's Manual          NGINX(8)

NAME
    nginx – HTTP and reverse proxy server, mail proxy server

SYNOPSIS
    nginx [-?hqTtVv] [-c file] [-g directives] [-p prefix]
          [-s signal]

DESCRIPTION
    nginx (pronounced “engine x”) is an HTTP and reverse proxy
    server, a mail proxy server, and a generic TCP/UDP proxy
    server. It is known for its high performance, stability,
    rich feature set, simple configuration, and low resource
    consumption.

    The options are as follows:

Manual page nginx(8) line 1 (press h for help or q to quit)
```

The options are as follows:

- ?, -h Print help.
- c file Use an alternative configuration file.
- g directives Set global configuration directives. See EXAMPLES for details.
- p prefix Set the prefix path. The default value is /usr/share/nginx.
- q Suppress non-error messages during configuration testing.
- s signal Send a signal to the master process. The argument signal can be one of: stop, quit, reopen, reload. The following table shows the corresponding system signals:

stop	SIGTERM
quit	SIGQUIT
reopen	SIGUSR1
reload	SIGHUP
- T Same as -t, but additionally dump configuration files to standard output.
- t Do not run, just test the configuration file. nginx checks the configuration file syntax and then tries to open files referenced in the configuration file.
- V Print the nginx version, compiler version, and configure script parameters.
- v Print the nginx version.

SIGNALS

The master process of nginx can handle the following signals:

SIGINT, SIGTERM	Shut down quickly.
SIGHUP	Reload configuration, start the new worker process with a new configuration, and gracefully shut down old worker processes.
SIGQUIT	Shut down gracefully.
SIGUSR1	Reopen log files.
SIGUSR2	Upgrade the nginx executable on the fly.
SIGWINCH	Shut down worker processes gracefully.

While there is no need to explicitly control worker processes normally, they support some signals too:

SIGTERM	Shut down quickly.
SIGQUIT	Shut down gracefully.
SIGUSR1	Reopen log files.

DEBUGGING LOG

To enable a debugging log, reconfigure nginx to build with debugging:

```
./configure --with-debug ...
```

and then set the debug level of the `error_log`:

```
error_log /path/to/log debug;
```

It is also possible to enable the debugging for a particular IP address:

```
events {
    debug_connection 127.0.0.1;
}
```

ENVIRONMENT

The `NGINX` environment variable is used internally by nginx and should not be set directly by the user.

FILES

`/run/nginx.pid`
Contains the process ID of nginx. The contents of this file are not sensitive, so it can be world-

ENVIRONMENT

The NGINX environment variable is used internally by nginx and should not be set directly by the user.

FILES

`/run/nginx.pid`
Contains the process ID of nginx. The contents of this file are not sensitive, so it can be world-readable.

`/etc/nginx/nginx.conf`
The main configuration file.

`/var/log/nginx/error.log`
Error log file.

EXIT STATUS

Exit status is 0 on success, or 1 if the command fails.

EXAMPLES

Test configuration file `~/mynginx.conf` with global directives for PID and quantity of worker processes:

```
nginx -t -c ~/mynginx.conf \  
-g "pid /var/run/mynginx.pid; worker_processes 2;"
```

SEE ALSO

Documentation at <http://nginx.org/en/docs/>.

For questions and technical support, please refer to <http://nginx.org/en/support.html>.

HISTORY

Development of nginx started in 2002, with the first public release on October 4, 2004.

AUTHORS

Igor Sysoev <igor@sysoev.ru>.

This manual page was originally written by Sergey A. Osokin <osa@FreeBSD.org.ru> as a result of compiling many nginx documents from all over the world.

BSD

December 5, 2019

BSD

Manual page nginx(8) line 91/135 (END) (press h for help or q to quit)

Step d – ps ax | grep nginx

```
[analyst@secOps ~]$ ps ax | grep nginx
```

```
[analyst@secOps ~]$ ps ax | grep nginx
2283 ?        Ss        0:00 nginx: master process /usr/sbin/nginx
2284 ?        S         0:00 nginx: worker process
3174 pts/2    S+        0:00 grep nginx
```

Step e – ls /etc/

```
[analyst@secOps ~]$ ls /etc/
```

```
[analyst@secOps ~]$ ls /etc/
adduser.conf      mime.types
alsa              mke2fs.conf
alternatives      modprobe.d
anacrontab         modules
apache2           modules-load.d
apm               mtab
apparmor          multipath
apparmor.d        nanorc
appport           netconfig
apt              netplan
bash.bashrc       network
bash_completion  networkd-dispatcher
bash_completion.d NetworkManager
bindresvport.blacklist networks
binfmt.d          newt
byobu             nftables.conf
ca-certificates   nginx
ca-certificates.conf nsswitch.conf
ca-certificates.conf.dpkg-old ntp.conf
cloud             oinkmaster.conf
compizconfig      openvswitch
console-setup     opt
cron.d            os-release
cron.daily        PackageKit
cron.hourly       pam.conf
cron.monthly      pam.d
crontab           papersize
cron.weekly       passwd
crypttab          passwd-
dbus-1            perl
dconf             pm
debconf.conf      pnm2ppa.conf
debian_version    polkit-1
default           pollinate
deluser.conf      ppp
depmod.d          profile
dhcp              profile.d
dictionaries-common protocols
dkms              pulse
dpkg              python3
e2scrub.conf      python3.10
emacs             rc0.d
environment       rc1.d
environment.d     rc2.d
ethertypes        rc3.d
```

firefox	rc4.d
fonts	rc5.d
freeradius	rc6.d
freetds	rc.local
fstab	rcS.d
ftpusers	resolv.conf
fuse.conf	rmt
gai.conf	rpc
ghostscript	rsyslog.conf
gimp	rsyslog.d
glvnd	scite
gnome	screenrc
groff	security
group	selinux
group-	sensors3.conf
grub.d	sensors.d
gshadow	services
gshadow-	shadow
gss	shadow-
gtk-2.0	shells
gtk-3.0	skel
guest-session	smi.conf
hdparm.conf	snort
host.conf	sos
hostname	sound
hosts	speech-dispatcher
hosts.allow	ssh
hosts.deny	ssl
ifplugd	subgid
inetd.conf	subgid-
init	subuid
init.d	subuid-
initramfs-tools	sudo.conf
inputrc	sudoers
iproute2	sudoers.d
issue	sudo_logsrvd.conf
issue.net	sysctl.conf
java-11-openjdk	sysctl.d
john	systemd
kernel	terminfo
landscape	timezone
ldap	tlp.d
ld.so.cache	tmpfiles.d
ld.so.conf	ubuntu-advantage
ld.so.conf.d	ucf.conf
legal	udev

```

libao.conf          udisks2
libaudit.conf       ufw
libblockdev         update-manager
libnl-3             update-motd.d
libpaper.d          update-notifier
libreoffice         UPower
lightdm             vdpau_wrapper.cfg
lighttpd            vim
locale.alias        vmware-tools
locale.gen          vsftpd.conf
localtime           vtrgb
logcheck            wgetrc
login.defs          wireshark
logrotate.conf      wpa_supplicant
logrotate.d         X11
lsb-release         xattr.conf
machine-id          xdg
magic               xinetd.conf
magic.mime          xinetd.d
manpath.config      xrdp
mate-settings-daemon zsh_command_not_found

```

Step f – ls -l /etc/nginx/

```
[analyst@secOps ~]$ ls -l /etc/nginx/
```

```

[analyst@secOps ~]$ ls -l /etc/nginx/
total 68
drwxr-xr-x 2 root root 4096 Nov 10 2022 conf.d
-rw-r--r-- 1 root root 2812 Feb 10 2023 custom_server.conf
-rw-r--r-- 1 root root 1125 Jul 27 2022 fastcgi.conf
-rw-r--r-- 1 root root 1055 Jul 27 2022 fastcgi_params
-rw-r--r-- 1 root root 2837 Jul 27 2022 koi-utf
-rw-r--r-- 1 root root 2223 Jul 27 2022 koi-win
-rw-r--r-- 1 root root 3957 Aug 2 2022 mime.types
drwxr-xr-x 2 root root 4096 Nov 10 2022 modules-available
drwxr-xr-x 2 root root 4096 Feb 10 2023 modules-enabled
-rw-r--r-- 1 root root 1447 Jul 27 2022 nginx.conf
-rw-r--r-- 1 root root 180 Jul 27 2022 proxy_params
-rw-r--r-- 1 root root 636 Jul 27 2022 scgi_params
drwxr-xr-x 2 root root 4096 Feb 10 2023 sites-available
drwxr-xr-x 2 root root 4096 Feb 10 2023 sites-enabled
drwxr-xr-x 2 root root 4096 Feb 10 2023 snippets
-rw-r--r-- 1 root root 664 Jul 27 2022 uwsgi_params
-rw-r--r-- 1 root root 3071 Jul 27 2022 win-utf

```

Step g – cat /etc/nginx/nginx.conf

```
[analyst@secOps ~]$ cat /etc/nginx/nginx.conf
```

```
[analyst@secOps ~]$ cat /etc/nginx/nginx.conf
user www-data;
worker_processes auto;
pid /run/nginx.pid;
include /etc/nginx/modules-enabled/*.conf;

events {
    worker_connections 768;
    # multi_accept on;
}

http {

    ##
    # Basic Settings
    ##

    sendfile on;
    tcp_nopush on;
    types_hash_max_size 2048;
    # server_tokens off;

    # server_names_hash_bucket_size 64;
    # server_name_in_redirect off;

    include /etc/nginx/mime.types;
    default_type application/octet-stream;

    ##
    # SSL Settings
    ##

    ssl_protocols TLSv1 TLSv1.1 TLSv1.2 TLSv1.3; # Dropping SSLv3, ref: P00DLE
    ssl_prefer_server_ciphers on;

    ##
    # Logging Settings
    ##

    access_log /var/log/nginx/access.log;
    error_log /var/log/nginx/error.log;

    ##
    # Gzip Settings
    ##
```

```
    gzip on;

    # gzip_vary on;
    # gzip_proxied any;
    # gzip_comp_level 6;
    # gzip_buffers 16 8k;
    # gzip_http_version 1.1;
    # gzip_types text/plain text/css application/json application/javascript text/xml application/xml application/xml+rss text/javascript;

    ##
    # Virtual Host Configs
    ##

    include /etc/nginx/conf.d/*.conf;
    include /etc/nginx/sites-enabled/*;
}

#mail {
#    # See sample authentication script at:
#    # http://wiki.nginx.org/ImapAuthenticateWithApachePhpScript
#
#    # auth_http localhost/auth.php;
#    # pop3_capabilities "TOP" "USER";
#    # imap_capabilities "IMAP4rev1" "UIDPLUS";
#
#    server {
#        listen     localhost:110;
#        protocol   pop3;
#        proxy      on;
#    }
#
#    server {
#        listen     localhost:143;
#        protocol   imap;
#        proxy      on;
#    }
#}
```

Step h - ls -l /var/log/

```
[analyst@secOps ~]$ ls -l /var/log/
```

```
[analyst@secOps ~]$ ls -l /var/log/
total 2228
-rw-r--r-- 1 root      root           0 Jan  2 00:00 alternatives.log
-rw-r--r-- 1 root      root      46356 Feb 10 2023 alternatives.log.1
drwxr-xr-x 2 root      root      4096 Jan  2 00:00 apt
-rw-r----- 1 syslog    adm       4450 Apr 12 22:46 auth.log
-rw-r----- 1 syslog    adm       5482 Apr 10 21:58 auth.log.1
-rw-r----- 1 syslog    adm       1169 Apr  2 12:17 auth.log.2.gz
-rw-r----- 1 syslog    adm        733 Jan 12 01:17 auth.log.3.gz
-rw-r----- 1 syslog    adm       2523 Jan  4 00:06 auth.log.4.gz
-rw-r--r-- 1 root      root     64549 Aug  9 2022 bootstrap.log
-rw-rw---- 1 root      utmp         0 Apr  6 23:07 btmp
-rw-rw---- 1 root      utmp         0 Jan  2 00:00 btmp.1
-rw-r--r-- 1 syslog    adm     96706 Feb 10 2023 cloud-init.log
-rw-r----- 1 root      adm      4935 Feb 10 2023 cloud-init-output.log
drwxr-xr-x 2 root      root      4096 Aug  3 2022 dist-upgrade
-rw-r----- 1 root      adm     45386 Apr 10 13:51 dmesg
-rw-r----- 1 root      adm     46130 Apr 10 13:43 dmesg.0
-rw-r----- 1 root      adm     14188 Jan 11 23:17 dmesg.1.gz
-rw-r----- 1 root      adm     14212 Dec 19 06:52 dmesg.2.gz
-rw-r----- 1 root      adm     14453 Feb 10 2023 dmesg.3.gz
-rw-r--r-- 1 root      root         0 Jan  2 00:00 dpkg.log
-rw-r--r-- 1 root      root    1131973 Apr  9 2025 dpkg.log.1
-rw-r--r-- 1 root      root     32288 Apr  9 2025 faillog
-rw-r--r-- 1 root      root     3838 Feb 10 2023 fontconfig.log
drwxr-xr-x 2 freerad   adm       4096 Dec 22 00:00 freeradius
-rw-r--r-- 1 root      root      1300 Apr 10 13:51 gpu-manager.log
drwxr-x-- 3 root      adm       4096 Feb 10 2023 installer
drwxr-sr-x+ 3 root     systemd-journal 4096 Feb 10 2023 journal
-rw-r----- 1 syslog    adm       3882 Apr 12 22:46 kern.log
-rw-r----- 1 syslog    adm     110357 Apr 12 10:37 kern.log.1
-rw-r----- 1 syslog    adm       1905 Apr  6 23:07 kern.log.2.gz
-rw-r----- 1 syslog    adm      13258 Jan 12 01:17 kern.log.3.gz
-rw-r----- 1 syslog    adm      12609 Jan  4 00:06 kern.log.4.gz
drwxr-xr-x 2 landscape landscape 4096 Feb 10 2023 landscape
-rw-rw---- 1 root      utmp    294628 Apr  9 2025 lastlog
drwxr-xr-x 2 root      root      4096 Apr 12 10:37 lightdm
-rw-r--r-- 1 root      root     24464 Jan 16 2023 logstash-tutorial.log
drwxr-xr-x 2 root      adm       4096 Dec 22 00:00 nginx
-rw-r--r-- 1 root      root         0 Jan 16 2023 nginx-logstash.log
drwxr-xr-x 2 ntp        ntp       4096 Feb 16 2022 ntpstats
drwxr-xr-x 2 root      root      4096 Apr 12 10:37 openvswitch
drwx----- 2 root      root      4096 Aug  9 2022 private
drwxr-s-- 2 snort       adm       4096 Feb 10 2023 snort
drwx----- 2 speech-dispatcher root     4096 May 23 2023 speech-dispatcher
-rw-r----- 1 syslog    adm     18462 Apr 12 22:46 syslog
-rw-r----- 1 syslog    adm     18462 Apr 12 22:46 syslog
-rw-r----- 1 syslog    adm    277607 Apr 12 10:37 syslog.1
-rw-r----- 1 syslog    adm     10749 Apr  6 23:07 syslog.2.gz
-rw-r----- 1 syslog    adm     22364 Jan 12 01:17 syslog.3.gz
-rw-r----- 1 syslog    adm     26470 Jan  4 00:06 syslog.4.gz
-rw-r--r-- 1 root      root         0 Jan  2 00:00 ubuntu-advantage.log
-rw-r--r-- 1 root      root      1583 Feb 10 2023 ubuntu-advantage.log.1
-rw-r--r-- 1 root      root      1150 Apr 12 17:06 ubuntu-advantage-timer.log
-rw-r--r-- 1 root      root      2990 Jan 11 23:37 ubuntu-advantage-timer.log.1
-rw-r--r-- 1 root      root     1041 Jan  1 23:37 ubuntu-advantage-timer.log.2.gz
drwxr-x-- 2 root      adm       4096 Feb 10 2023 unattended-upgrades
-rw-r----- 1 root      root         678 Feb 10 2023 vboxadd-install.log
-rw-r--r-- 1 root      root         157 Feb 10 2023 vboxadd-setup.log
-rw-r----- 1 root      adm         0 Apr  6 23:07 vsftpd.log
-rw-r----- 1 root      root     1120 Apr  2 12:00 vsftpd.log.1
-rw-rw---- 1 root      utmp     9216 Apr 10 12:35 wtmp
-rw-r--r-- 1 root      root    36977 Apr 12 23:03 Xorg.0.log
-rw-r--r-- 1 root      root    23681 Apr 10 11:43 Xorg.0.log.old
```

Step i – sudo ls -l /var/log/nginx

```
[analyst@secOps ~]$ sudo ls -l /var/log/nginx
```

```
[analyst@secOps ~]$ ls -l /var/log/nginx
total 4
-rw-r----- 1 www-data adm 0 Feb 10 2023 access.log
-rw-r----- 1 www-data adm 0 Dec 22 00:00 error.log
-rw-r----- 1 www-data adm 78 Feb 10 2023 error.log.1
```

Part 3 – Monitoring Log Files in Real Time

Step 1a – sudo tail /var/log/nginx/access.log

```
[analyst@secOps ~]$ sudo tail /var/log/nginx/access.log
```

```
[analyst@secOps ~]$ sudo tail /var/log/nginx/access.log
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET / HTTP/1.1" 200 147 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:57 -0400] "GET / HTTP/1.1" 200 612 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:58 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:38:48 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:56:14 -0400] "afdfdasdf" 400 157 "-" "-"
```

Step 1b – sudo tail -n 5 /var/log/nginx/access.log

```
[analyst@secOps ~]$ sudo tail -n 5 /var/log/nginx/access.log
```

```
[analyst@secOps ~]$ sudo tail -n 5 /var/log/nginx/access.log
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:57 -0400] "GET / HTTP/1.1" 200 612 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:58 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:38:48 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:56:14 -0400] "afdfdasdf" 400 157 "-" "-"
```

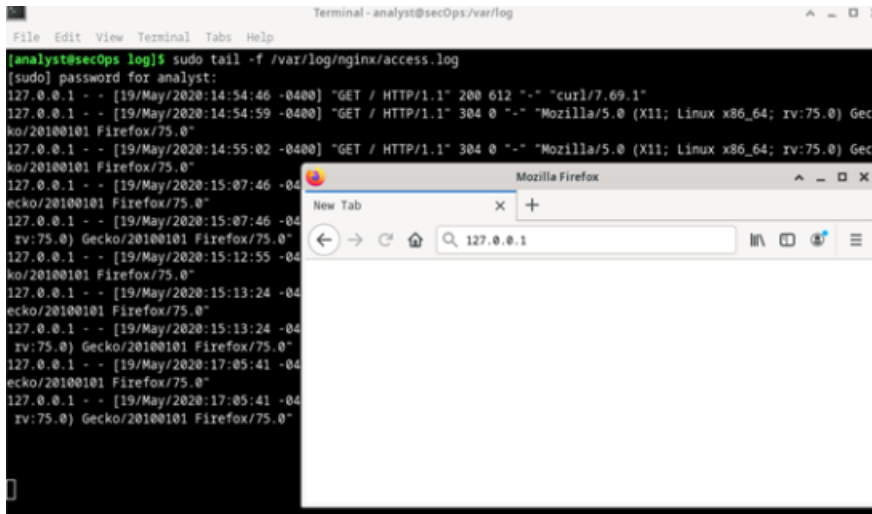
Step 1c – sudo tail -f (Real-Time Monitor)

```
[analyst@secOps ~]$ sudo tail -f /var/log/nginx/access.log
```

```
[analyst@secOps ~]$ sudo tail -f /var/log/nginx/access.log
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET / HTTP/1.1" 200 147 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:12:07:42 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:57 -0400] "GET / HTTP/1.1" 200 612 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:37:58 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:38:48 -0400] "GET /favicon.ico HTTP/1.1" 404 153 "-" "Mozilla/5.0 (X11; Linux x86_64; rv:75.0) Gecko/20100101 Firefox/75.0"
127.0.0.1 - - [01/Sep/2020:14:56:14 -0400] "afdfdasdf" 400 157 "-" "-"
```

Step 1d & e – Browsing to 127.0.0.1 while tail -f is running

A web browser was opened and navigated to 127.0.0.1. The nginx server served the request and new entries appeared in real time at the bottom of the terminal window running tail -f.



Step 2a - journalctl (no options)

```
[analyst@secOps ~]$ journalctl
Feb 10 21:25:19 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] AppArmor>
Feb 10 21:25:19 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Activati>
Feb 10 21:25:19 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Successf>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:27:44 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
Feb 10 21:30:06 labvm dbus-daemon[72440]: [session uid=1002 pid=72438] Reloaded>
-- Boot 13212d852df649f1a2bc1e52d8e8b87b --
Apr 10 12:35:36 labvm systemd-xdg-autostart-generator[1199]: Configuration file>
Apr 10 12:35:36 labvm systemd[1192]: Queued start job for default target Main U>
Apr 10 12:35:36 labvm systemd[1192]: Created slice User Application Slice.
Apr 10 12:35:36 labvm systemd[1192]: Created slice User Core Session Slice.
Apr 10 12:35:36 labvm systemd[1192]: Reached target Paths.
Apr 10 12:35:36 labvm systemd[1192]: Reached target Timers.
Apr 10 12:35:36 labvm systemd[1192]: Starting D-Bus User Message Bus Socket...
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG network certificate man>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent and>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent and>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent (ss>
Apr 10 12:35:36 labvm systemd[1192]: Listening on GnuPG cryptographic agent and>
lines 1-22
```

Answer - How to run journalctl and see ALL log entries?

Run journalctl as root using sudo:

```
sudo journalctl
```

This grants access to all journal entries, including those belonging to other users and the system. Alternatively, the analyst user could be added to the systemd-journal, adm, or wheel group, which would allow journalctl to display all messages without sudo.

Step 2b – sudo journalctl -b (boot entries)

```
[analyst@secOps ~]$ sudo journalctl -b
Apr 10 13:51:25 labvm kernel: Linux version 5.15.0-60-generic (buildd@lcy02-amd>
Apr 10 13:51:25 labvm kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-5.15.0-60->
Apr 10 13:51:25 labvm kernel: KERNEL supported cpus:
Apr 10 13:51:25 labvm kernel:   Intel GenuineIntel
Apr 10 13:51:25 labvm kernel:   AMD AuthenticAMD
Apr 10 13:51:25 labvm kernel:   Hygon HygonGenuine
Apr 10 13:51:25 labvm kernel:   Centaur CentaurHauls
Apr 10 13:51:25 labvm kernel:   zhaoxin   Shanghai
Apr 10 13:51:25 labvm kernel: [Firmware Bug]: TSC doesn't count with P0 frequen>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x001: 'x87 flo>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x002: 'SSE reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x004: 'AVX reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: xstate_offset[2]:  576, xstate_sizes[2]:>
Apr 10 13:51:25 labvm kernel: x86/fpu: Enabled xstate features 0x7, context siz>
Apr 10 13:51:25 labvm kernel: signal: max sigframe size: 1776
Apr 10 13:51:25 labvm kernel: BIOS-provided physical RAM map:
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000000000>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000000009fc00-0x00000000000000>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x000000000000f0000-0x0000000000000f>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000000100000-0x000000000007ffe>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0000000007fff>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0>
lines 1-22
```

Step 2c – sudo journalctl -b -2 (two boots ago)

```
[analyst@secOps ~]$ sudo journalctl -b -2
Apr 10 13:51:25 labvm kernel: Linux version 5.15.0-60-generic (buildd@lcy02-amd>
Apr 10 13:51:25 labvm kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-5.15.0-60->
Apr 10 13:51:25 labvm kernel: KERNEL supported cpus:
Apr 10 13:51:25 labvm kernel:   Intel GenuineIntel
Apr 10 13:51:25 labvm kernel:   AMD AuthenticAMD
Apr 10 13:51:25 labvm kernel:   Hygon HygonGenuine
Apr 10 13:51:25 labvm kernel:   Centaur CentaurHauls
Apr 10 13:51:25 labvm kernel:   zhaoxin   Shanghai
Apr 10 13:51:25 labvm kernel: [Firmware Bug]: TSC doesn't count with P0 frequen>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x001: 'x87 flo>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x002: 'SSE reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: Supporting XSAVE feature 0x004: 'AVX reg>
Apr 10 13:51:25 labvm kernel: x86/fpu: xstate_offset[2]:  576, xstate_sizes[2]:>
Apr 10 13:51:25 labvm kernel: x86/fpu: Enabled xstate features 0x7, context siz>
Apr 10 13:51:25 labvm kernel: signal: max sigframe size: 1776
Apr 10 13:51:25 labvm kernel: BIOS-provided physical RAM map:
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000000000000-0x00000000000000>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000000009fc00-0x00000000000000>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x000000000000f0000-0x0000000000000f>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000000100000-0x000000000007ffe>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x0000000007fff0000-0x0000000007fff>
Apr 10 13:51:25 labvm kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec0>
lines 1-22
```

Step 2d – sudo journalctl --list-boots

```
[analyst@secOps ~]$ sudo journalctl --list-boots  
[analyst@secOps ~]$ sudo journalctl --list-boots  
-4 cc7dd27d5a8a4df1a4d3457c30f9dd4b Fri 2023-02-10 21:10:50 UTC-Fri 2023-02-10 >  
-3 a9b24cc963364eceb493c7c47117432e Fri 2025-12-19 08:52:06 UTC-Wed 2026-01-07 >  
-2 fcc2db4f1c554d52a0ea958c6e8b6edc Mon 2026-01-12 01:17:08 UTC-Mon 2026-04-06 >  
-1 c285fe5cf8224dc38fd716f36db7a9e0 Fri 2026-04-10 13:43:06 UTC-Fri 2026-04-10 >  
0 13212d852df649f1a2bc1e52d8e8b87b Fri 2026-04-10 13:51:25 UTC-Sun 2026-04-12 >  
lines 1-5/5 (END)
```

Step 2e – Filtering by time (--since)

```
[analyst@secOps ~]$ sudo journalctl --since '2 hours ago'  
[analyst@secOps ~]$ sudo journalctl --since '1 day ago'  
Apr 12 22:46:27 labvm kernel: 22:46:27.441267 timesync vgsvcTimeSyncWorker: Rad>  
Apr 12 22:46:27 labvm CRON[2897]: pam_unix(cron:session): session opened for us>  
Apr 12 22:46:27 labvm systemd[1]: Condition check resulted in Run anacron jobs >  
Apr 12 22:46:27 labvm CRON[2900]: (root) CMD ( [ -x /etc/init.d/anacron ] && if >  
Apr 12 22:46:27 labvm CRON[2897]: pam_unix(cron:session): session closed for us>  
Apr 12 23:06:34 labvm systemd[1]: Starting Ubuntu Advantage Timer for running r>  
Apr 12 23:06:35 labvm systemd[1]: ua-timer.service: Deactivated successfully.  
Apr 12 23:06:35 labvm systemd[1]: Finished Ubuntu Advantage Timer for running r>  
Apr 12 23:07:34 labvm systemd-resolved[2489]: Clock change detected. Flushing c>  
Apr 12 23:07:40 labvm ntpd[648]: Soliciting pool server 196.10.52.58  
Apr 12 23:07:41 labvm ntpd[648]: Soliciting pool server 102.205.240.0  
Apr 12 23:09:18 labvm sudo[3098]: analyst : TTY=pts/0 ; PWD=/home/analyst ; US>  
Apr 12 23:09:18 labvm sudo[3098]: pam_unix(sudo:session): session opened for us>  
Apr 12 23:09:18 labvm sudo[3098]: pam_unix(sudo:session): session closed for us>  
Apr 12 23:11:47 labvm sudo[3103]: analyst : TTY=pts/0 ; PWD=/home/analyst ; US>  
Apr 12 23:11:47 labvm sudo[3103]: pam_unix(sudo:session): session opened for us>  
Apr 12 23:11:47 labvm sudo[3103]: pam_unix(sudo:session): session closed for us>  
Apr 12 23:12:19 labvm sudo[3106]: analyst : TTY=pts/0 ; PWD=/home/analyst ; US>  
Apr 12 23:12:19 labvm sudo[3106]: pam_unix(sudo:session): session opened for us>  
Apr 12 23:12:19 labvm sudo[3106]: pam_unix(sudo:session): session closed for us>  
Apr 12 23:13:31 labvm wnck-applet[1610]: Negative content width -1 (allocation >  
Apr 12 23:13:31 labvm wnck-applet[1610]: Negative content width -1 (allocation >  
lines 1-22
```

Step 2f – sudo journalctl -u nginx.service

```
[analyst@secOps ~]$ sudo journalctl -u nginx.service  
[analyst@secOps ~]$ sudo journalctl -u nginx.service  
Feb 10 21:29:12 labvm systemd[1]: Starting A high performance web server and a >  
Feb 10 21:29:12 labvm systemd[1]: Started A high performance web server and a r>  
Feb 10 21:31:17 labvm systemd[1]: Stopping A high performance web server and a >  
Feb 10 21:31:17 labvm systemd[1]: nginx.service: Deactivated successfully.  
Feb 10 21:31:17 labvm systemd[1]: Stopped A high performance web server and a r>  
lines 1-5/5 (END)
```

Step 2g – sudo journalctl -f (real-time follow)

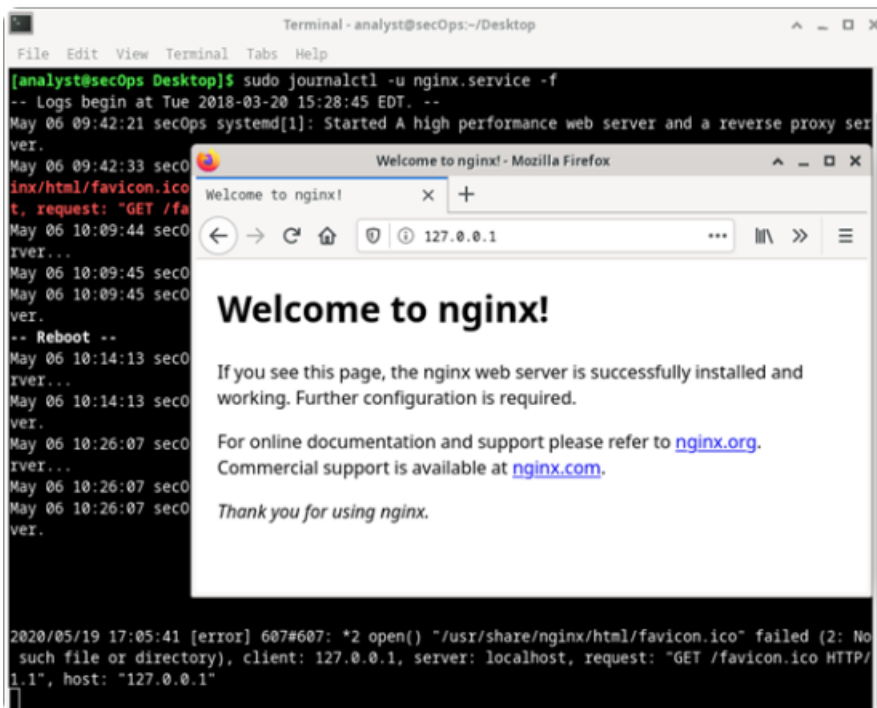
```
[analyst@secOps ~]$ sudo journalctl -f
```

```
[analyst@secOps ~]$ sudo journalctl -f
Apr 12 23:52:54 labvm sudo[3355]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
Apr 12 23:53:53 labvm sudo[3355]: pam_unix(sudo:session): session closed for use
r root
Apr 12 23:54:27 labvm sudo[3359]: analyst : TTY=pts/0 ; PWD=/home/analyst ; USE
R=root ; COMMAND=/usr/bin/journalctl --since 2 hours ago
Apr 12 23:54:27 labvm sudo[3359]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
Apr 12 23:55:29 labvm sudo[3359]: pam_unix(sudo:session): session closed for use
r root
Apr 12 23:56:06 labvm sudo[3363]: analyst : TTY=pts/0 ; PWD=/home/analyst ; USE
R=root ; COMMAND=/usr/bin/journalctl -u nginx.service
Apr 12 23:56:06 labvm sudo[3363]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
Apr 12 23:57:13 labvm sudo[3363]: pam_unix(sudo:session): session closed for use
r root
Apr 12 23:57:28 labvm sudo[3368]: analyst : TTY=pts/0 ; PWD=/home/analyst ; USE
R=root ; COMMAND=/usr/bin/journalctl -f
Apr 12 23:57:28 labvm sudo[3368]: pam_unix(sudo:session): session opened for use
r root(uid=0) by (uid=1002)
```

Step 2h & i – journalctl -u nginx.service -f + browser test

```
[analyst@secOps ~]$ sudo journalctl -u nginx.service -f
```

A browser was opened and directed to 127.0.0.1. A missing favicon.ico error appeared in real time in the journalctl output.



Reflection & Key Takeaways

- Log files are essential for troubleshooting system and application events. They provide a historical record that allows analysts to trace what happened and when.
- Log file locations follow conventions (e.g., /var/log/), but are ultimately chosen by the developer. Always consult documentation first.
- When documentation is unavailable, a combination of web research and direct system investigation (manual pages, configuration files, directory listings) is effective.
- Correct clock synchronization (NTP) is critical. Inaccurate timestamps make it very difficult to correlate and trace events across multiple systems.
- Events from different sources are often analyzed together. Consistent, accurate timestamps allow log entries from the OS, web server, firewall, and other services to be correlated.
- The tail -f command and journalctl -f are powerful real-time monitoring tools. They allow analysts to observe live system and application behavior without waiting for a static log dump.

Commands Reference Summary

Command	Purpose
cat /var/log/logstash-tutorial.log	Display entire log file at once
sudo more /var/log/messages	Page through a log file (root required)
man nginx	Read nginx manual page for log file hints
ps ax grep nginx	Verify nginx process is running
ls /etc/ & ls -l /etc/nginx/	Locate nginx configuration directory and files
cat /etc/nginx/nginx.conf	Read nginx configuration to find log paths
ls -l /var/log/	List all log files and directories
sudo ls -l /var/log/nginx	List nginx-specific log files (root required)
sudo tail /var/log/nginx/access.log	Display last 10 lines of nginx access log
sudo tail -n 5 /var/log/nginx/access.log	Display last 5 lines
sudo tail -f /var/log/nginx/access.log	Real-time log monitoring (follow mode)
journalctl	Display all journald log entries
sudo journalctl -b	Display current-boot journal entries
sudo journalctl -b -2	Display entries from two boots ago
sudo journalctl --list-boots	List all recorded boot sessions
sudo journalctl --since '2 hours ago'	Filter entries by time range
sudo journalctl -u nginx.service	Show entries for a specific service unit
sudo journalctl -f	Real-time journal monitoring
sudo journalctl -u nginx.service -f	Real-time monitoring for nginx service